

SUMMARY

I aspire to be a global leader in respiratory epidemiology. As a 3 yr experienced clinical pharmacist, I have a strong interest in respiratory epidemiology, statistical methods and evidence synthesis. So far, I produced **31** publications, including **17** as the first author, including **Lancet Respiratory Medicine**, **JAMA internal medicine** and **European Respiratory Journal** and secured **four highly competitive research grants**. I built my independent research networks in other UK universities (UCL, Edinburgh, Aberdeen, Southampton) and other countries (Finland, Sweden, South Korea, Spain, Singapore and Ghana).

EMPLOYMENT

Postdoctoral Researcher, Imperial College London, May 2023- May 2027

Skills: R, STATA, Data wrangling, Data cleaning, Self-controlled case series, Inverse probability of treatment weighting, Propensity score, Negative control analyses, Regression models, Target trial emulation, Mendelian randomisation

Research: Led/leading **7** respiratory epidemiology research using big health data, e.g., Clinical Practice Research Datalink (CPRD), UK Biobank (UKB) and a birth cohort (SEATON), **3** evidence synthesis and **1** AI-driven research

- (1) Metformin for asthma attacks in adults (CPRD)
 - Published in *JAMA internal medicine*
 - Selected *one of the six major studies set to influence asthma management* in 2025 American Thoracic Society
- (2) Oral contraceptives for asthma attacks in women (CPRD)
 - Published in *ERJ Open Research* and 53 press.
- (3) Pregnancy and asthma attacks (CPRD)
 - Explored asthma attacks patterns pre/during/after pregnancy and modifiable risk factors
 - Published in *European Respiratory Journal*
 - Presented at *2025 European Respiratory Society Congress* in Amsterdam
- (4) Hormonal replacement therapy for asthma exacerbation in women (CPRD, published in ERJOR)
- (5) Metformin for COPD and IPF (CPRD, ongoing)
- (6) Causal association between Polycystic ovary syndrome and asthma development (UKB, ongoing)
- (7) Evidence synthesis: Effects of diabetes medications on adverse lung outcomes
- (8) Ultra-processed food and asthma development in children (SEATON cohort) (Independent project with a University of Edinburgh team)
- (9) Evidence synthesis: Ultra-processed food (prenatal and postnatal) and asthma/allergy development (Independent project with a South Korean research team)
- (10) Evidence synthesis: Spacers and valved holding chambers used for delivering inhaled medication (Independent project with a Finnish research team)
- (11) Designing AI-assisted priority setting and risk of bias assessment for evidence synthesis (Independent project with a Southampton research team and WHO)

ACADEMIC SERVICE

Associate Editor, BMJ Open Respiratory Research, 2024-Present, Guest Editor (Special collection 'asthma management')

Associate Editor, ERJ Open Research, 2024-Present

FUNDING AND AWARDS: To date, I have secured **four** competitive grants during my postdoc.

Project Grants: Principal Investigator

- Early Career Researcher Fellowship , Asthma + Lung UK (£100K, 2024) ECSG24\8; Feb 2025 – Jun 2026: Antidiabetic medication for COPD and IPF

- NIHR Imperial BRC Project Award (£15K, 2025, PI): Causal relationship between Polycystic ovary syndrome and asthma development using Mendelian Randomisation
- National Heart and Lung Institute (NHLI) Pilot Award (£880, 2025)
- University of Edinburgh Student Experience Grant (£5K, 2023): Project: Exploring the role of culture and language in gaining "Masterness" supporting international students

Travel Grants: NHLI Travel Award*2 (2023,2025), International Society for Pharmacoepidemiology (2024)

Awards:

- Annual recognition award for outstanding contribution to the student experience (Usher Institute, 2022)
- Annual recognition award for most outstanding team (Usher institute, 2021)
- Award of recognition (Korean Pharmaceutical Association, South Korea, 2016)

TEACHING & SUPERVISION

I was awarded **AFHEA** from Advance HE (2025)

- Taught study designs for MSc in Cardiovascular and Respiratory Healthcare, NHLI, Imperial College
- Mentored 13 MSc students as Personal Tutor (Imperial)
- Supervised 13 BSc (Pharmacy, Medicine), 7 Master's (Edinburgh) and 1 PhD student (Imperial): Many students had 1st in the project and two students have presented at national conferences and co-author on a publication. Students demonstrated my clear learning plan, timely and thorough feedback, open communications and empowerment to students.
- Mentoring postgraduate students at Kwame Nkrumah University of Science and Technology, in Ghana (2025-Present)

EVIDENCE OF ESTEEM

Committee activities

- International Society for Pharmacoepidemiology (ISPE): Education, Publication (2023-Present)
Organise seminars and provide educational materials
- **Co-Chair** of NHLI Postdoc and Research fellow committee (2023-Present)
Review NHLI grant applications and organise events (e.g. NHLI Career day) for early career researchers

Organising a conference or symposium

- UNCOVER Annual conference (2022): as part of the Applied Evidence Synthesis group at Usher Institute, University of Edinburgh
- 3-day webinar for evidence synthesis (2022) at Universiti Malaya in Malaysia
- ISPE annual meeting (2024): I was involved in environmental initiatives for the annual conference
- 1-day meta-analysis webinar for postgraduate students at Kwame Nkrumah University of Science and Technology, in Ghana (2026)

Membership of learned societies

I am an active member of British Thoracic Society (2023-Present) and European Respiratory Society (2025-Present)

EDUCATION

- **PhD Research (Major: Population Health Sciences), Usher institute, University of Edinburgh, UK**
funded by the Asthma UK Centre for Applied Research
Supervisor: Steven Cunningham, Steff Lewis, Steve Turner
Project: The effectiveness of oral corticosteroids for preschool wheeze: Individual patient data meta-analysis (No correction)
- **Master's Program (Major: Public Health), Usher institute, University of Edinburgh, UK**
- **Master's Program (Major: Clinical Pharmacy), Department of Pharmacy, Dankook University, South Korea**
- **Doctor of Pharmacy (PharmD) College of Pharmacy, Dankook University, South Korea**

PUBLICATIONS (<https://scholar.google.com/citations?user=tU-CZS8AAAAJ&hl=en>)

I have **31** peer-reviewed articles (including **17** first authors, **2** editorials) and **8** manuscripts under review

SELECTED PEER-REVIEW PUBLICATIONS (Total 31, selected papers as first or corresponding author*)

1. **Lee B**, Anpalagan H, Tan T, Wong E, Bloom C. Hormone replacement therapy and the risk of asthma attacks: population-based cohort study. *ERJ Open Res.* 2026. 01599-2025
2. **Lee B**, Wong E, Tan T, Rupani H, Bloom CI. Pregnancy, asthma and exacerbations: a population-based cohort. *Eur Respir J.* 2025;66(6).
3. **(Editorial) Lee B**, Balsells E, Shi T. Bridging the Gap: Advancing Equity in Improving Childhood Lower Respiratory Infections. *Paediatr Perinat Epidemiol.* 2025;39(8):657-9.
4. **Lee B**, Trusinska D, Ferdous S, et al. Real-world effectiveness and safety of nirsevimab, RSV maternal vaccine and RSV vaccines for older adults: a living systematic review and meta-analysis. *Thorax.* 2025;80(11):838-48.
5. **(Editorial) Lee B**, Song WJ. Redefining chronic cough as a disease: why patient registries matter. *ERJ Open Res.* 2025;11(5).
6. **Lee B**, Turner SW, Hine J, et al. Consensus outcomes between health professionals and parents for oral corticosteroids in treating preschool wheeze: a multi-national survey and nominal group technique study. *Arch Dis Child.* 2025;110(7):521-7.
7. **Lee B**, Man KKC, Wong E, Tan T, Sheikh A, Bloom CI. Antidiabetic medication and asthma attacks. *JAMA Intern Med.* 2025;185(1):16-25.
8. **Lee B**, Fard ARR, Wong E, Tan T, Bloom CI. Oral contraceptives and the risk of asthma attacks: a population-based cohort study. *ERJ Open Res.* 2025;11(5).
9. **Lee B***, Turner S, Borland ML, et al. Efficacy of Oral Corticosteroids for Acute Preschool Wheeze: A Systematic Review and Individual Participant Data Meta-Analysis of Randomised Clinical Trials. *Lancet Respir Med.* 2024. **(*corresponding author)**
10. **Lee B**, Krishan P, Goodwin L, et al. Impact of COVID-19 mitigations on anxiety and depression amongst university students: A systematic review and meta-analysis. *J Glob Health.* 2023;13:06035.
11. **Lee B**, Ashcroft T, Agyei-Manu E, et al. Clinical features of COVID-19 for integration of COVID-19 into influenza surveillance: A systematic review. *J Glob Health.* 2022;12:05012.
12. **Lee B**, Lewis G, Agyei-Manu E, et al. Risk of serious COVID-19 outcomes among adults and children with moderate-to-severe asthma: a systematic review and meta-analysis. *Eur Respir Rev.* 2022;31(166).
13. **Lee B***, Ibrahim SA, Zhang T. Mobile apps leveraged in the COVID-19 pandemic in East and South-East Asia: review and content analysis. *JMIR mHealth uHealth.* 2021;9(11):e32093. **(*corresponding author)**
14. **Lee B**, Lee YJ. Evaluation of medication use and pharmacy services for visually impaired persons: Perspectives from both visually impaired and community pharmacists. *Disabil Health J.* 2019;12(1)

SELECTED MANUSCRIPTS UNDER REVIEW (Total, 8 studies are under review: 1 as first author, 1 as corresponding author and 7 as co-author)

1. **Lee B**, Kotikoski A, Ojanperä L, et al. Spacers and valved holding chambers used for delivering inhaled medication – a scoping review of the quantitative and qualitative evidence from in vitro studies. 2026
2. McQuaid F, Skafida V, McQuillan R, **Lee B***. Screen-use and emotional, behavioural and social development in children: a cross-sectional and longitudinal cohort analysis. 2026 (under review) (* corresponding author, JAMA Network Open)

Other **7** studies were from my works in collaboration with research teams at Univ. of Edinburgh, Univ. of Singapore (Singapore), Univ. of Tampere (Finland) and National Heart Hospital in Zambia (Zambia)

CONFERENCES

I have presented my research at national or international prestigious societies via oral or poster presentation: Total **18** abstracts: **13** in European Research Society, **2** in Cochrane Colloquium, **1** in British Thoracic Society, **1** in Pharmacoepidemiology and **1** in Public Health)